Cognitive-Oriented Reasoning Engine for Procrastination Analysis

Angel Sharma, Kanak Jain, Anika Verma, Surbhi Vijh

Department of Computer Science, Amity University Noida, Uttar Pradesh, India

Abstract

Procrastination is a complex behavior impacted by emotions, motivation, and environmental factors, often resulting in lower productivity and mental health challenges. This paper presents the CORE-AI framework, a modular, multi-layer inference architecture designed to move beyond simple text classification toward structured cognitive reasoning. The system utilizes a conversational web agent to analyze real-time dialogue and identify root causes such as perfectionism, fear of failure, and dopamine-driven impulses. The core infrastructure employs a fine-tuned Two-Head BERT model to detect mental states related to anxiety and depression. For behavioral analysis, the system treats root causes as a multi-label classification task, identifying overlapping drivers like environmental distractions and lack of interest. A Psychological Risk Fusion Engine integrates these indicators to estimate structured risk levels while maintaining historical continuity to reduce bias from isolated instances. To bridge the gap between analysis and action, CORE-AI features an Automated Report Generation Module that delivers comprehensive psychological summaries directly to the user via email. Furthermore, the framework generates a Personalized Planner, which suggests specific behavioral tasks and mitigation strategies (e.g., Pomodoro techniques for distraction) tailored to the identified root causes. The interface, built with Streamlit, ensures natural dialogue and explainable outputs through an Explainable Decision Layer. This framework connects behavioral science with AI, encouraging self-awareness through a scalable, inference-only deployment model that mirrors the complex nature of procrastination.

Proposed Technique

The proposed system, CORE-AI, is a modular and scalable framework designed for real-time psychological risk assessment. It integrates conversational empathy with a multi-stage AI pipeline to process and analyse user dialogue continuously. The architecture leverages a hybrid approach, utilizing deep learning for contextual emotional screening alongside classical machine learning to identify behavioral triggers. By aggregating these insights through a Psychological Risk Fusion Engine, the framework ensures accurate, explainable, and continuous cognitive risk prediction in real-world environments.

A. Project Overview:

The proposed system, CORE-AI, is a modular and scalable framework designed for real-time psychological risk assessment and behavioral intervention. It integrates conversational empathy via an intelligent web agent with a multi-stage AI pipeline to process and analyze user dialogue continuously. Architecture leverages a hybrid approach, utilizing deep learning for contextual emotional screening alongside classical machine learning to identify behavioral triggers. By aggregating these insights through a Psychological Risk Fusion Engine, the framework ensures accurate and explainable cognitive risk prediction. To translate these insights into action, the system features an Automated Report Generation module that delivers structured psychological summaries via email, paired with a Personalized Planner that generates specific behavioral tasks tailored to the identified root causes. This end-to-end deployment ensures a continuous, privacy-preserving feedback loop in real-world environments.

B. Model Architecture:

The system utilizes a modular AI pipeline combining deep learning and classical machine learning for comprehensive analysis:

Two-Head BERT Model: A shared deep-learning BERT encoder with two output heads (Depression and Anxiety). It uses 80th-percentile (p80) pooling and temperature scaling (T=1.5) to accurately capture contextual emotional nuances from dialogue.

Root-Cause Classifier: A classical machine-learning module utilizing a TF-IDF vectorizer and One-vs-Rest Logistic Regression for multi-label prediction of behavioral triggers (Perfectionism, Fear of Failure, Lack of Interest, Environmental Distraction, Dopamine Addiction).

Psychological Risk Fusion Engine: A decision-based logic layer that integrates disorder probabilities and behavioral drivers into a single, unified cognitive risk framework, supported by an explainable decision layer.

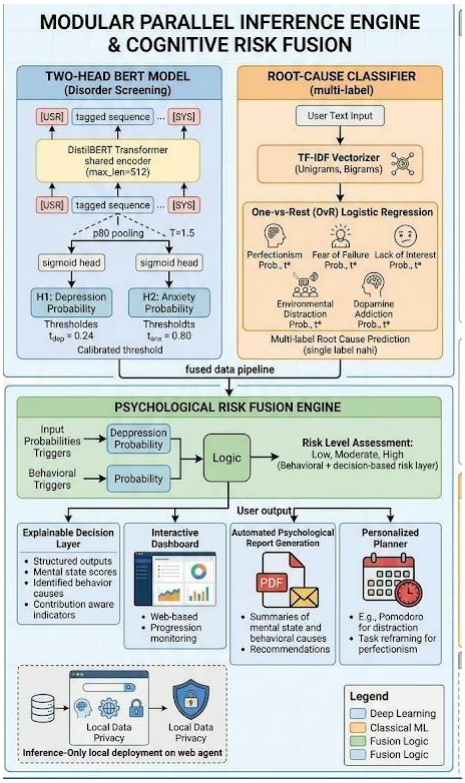

Fig.1: Modular AI System: Multi-Stage Reasoning Flow

C. Model Training and Evaluation:

The models were trained using balanced disorder screening datasets and a custom behavioral root-cause dataset. The deep learning module utilized binary cross-entropy with logits, applying Focal Loss (gamma=2.0, alpha=0.75) to minimize false positives, and was optimized using AdamW. The classical components were trained using class-balanced weights. Decision thresholds were dynamically tuned and calibrated individually per class on validation data to maximize F1 scores (e.g., t*=0.24 for depression to catch muted cues, and t*=0.80 for anxiety). Evaluations were carried out using multiple metrics including ROC-AUC, PR-AUC, Precision, Recall, and F1-score, ensuring robust performance and real-time stability.

Experimental Results

The CORE-AI system was evaluated using a held-out test set, where the transition to a conversation-aware architecture with 80th-percentile (p80) pooling and temperature scaling (T=1.5) yielded the most significant gains in stability. The Two-Head BERT model achieved high accuracy (0.9817 for Depression and 0.9906 for Anxiety), with optimized thresholds (t*=0.24$ and 0.80, respectively) ensuring precise sensitivity to muted emotional cues.

Model Head	Accuracy	Precision	Recall	F1	t*
Depression	0.9817	0.9866	0.9761	0.9813	0.24
Anxiety	0.9906	0.9893	0.9919	0.9906	0.80

Table.1: Final Model Performance for Disorder Causes

Model Head	Average Precision	Precision	Recall	F1	t*
Perfectionism	0.925	1.000	0.833	0.909	0.90
Fear of Failure	0.889	0.800	1.000	0.899	0.20
Lack of Interest	0.857	0.750	1.000	0.857	0.40
Environmental Distraction	0.800	0.769	0.833	0.800	0.05
Dopamine Addiction	0.727	0.571	1.000	0.727	0.20

Table.2: Final Model Performance for Non- Disorder Causes

The Multi-label Root-Cause Classifier demonstrated excellent precision in identifying behavioral drivers like Environmental Distraction and Dopamine Addiction through its TF-IDF → One-vs-Rest framework. The Psychological Risk Fusion Engine successfully integrated these signals into a single cognitive risk estimation (Low, Moderate, High), mitigating the impact of atypically high language scores from single occurrences.

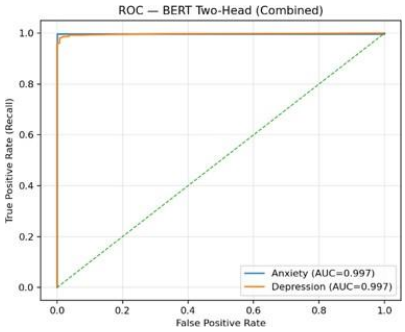

Fig.2: ROC- Bert Two-Head Model

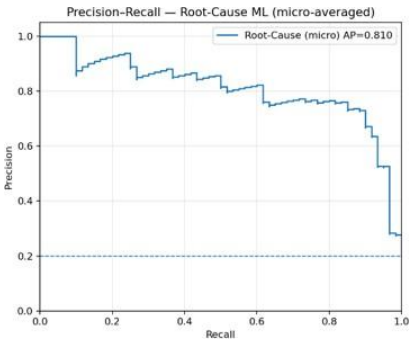

Fig.3: Precision–Recall -Root-Cause ML (micro-averaged)

Validation via the Streamlit-based dashboard confirmed real-time inference capabilities with minimal latency and accurate Automated Report Generation. By utilizing an Explainable Decision Layer, the system provided structured, contribution-aware outputs, allowing users to see the reasoning structure behind their results and fostering trust through transparency.

Conclusions

The CORE-AI framework unifies mental state detection with multi-label behavioral modelling. By integrating BERT transformers with ML classifiers (TF-IDF + Logistic Regression), the system provides a comprehensive view of why users delay tasks. The inclusion of cognitive risk fusion ensures results are interpretable insights rather than raw predictions. Results show the system effectively distinguishes psychological and behavioral causes with high generalizability. Prioritizing transparency and a privacy-focused architecture, CORE-AI demonstrates how responsible AI can bridge technology and psychology to help users understand habits and improve productivity through structured, evidence-based insight.